Addis Ababa Science and Technology University

Mathematics for Natural Sciences

By

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Main Objectives of this Course

Chapter One outlines

- Know the basic concepts of Propositional Logic
- Understand the concept of set.

Chapter One

Propositional Logic and Set Theory

What is Proposition?

Defn: A proposition (or statement) is a sentence which has a truth value (either True or False but not both).

Example Consider the following sentences.

a. 2 is an even number.

b. A triangle has four sides.

c. May God bless you!

d. Give me that book.

e. What is your name?

here a & b is a proposition and c , d and e are not proposition

Logical connectives

1. Conjunction

When two propositions are joined with the connective "**and**," the proposition formed is a logical ***conjunction*** **and denoted by " ^". is written: $p \wedge q$, read as "p and q ," or "p conjunction q ".**

P	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example Consider the following propositions:

p : 3 is an odd number. (True)

q : 27 is a prime number. (False)

$p \wedge q$: 3 is an odd number and 27 is a prime number. (False)

2. Disjunction

When two propositions are joined with the connective "**or**," the proposition formed is called a logical ***disjunction***. **"or" and denoted "$\vee$ ". So, the logical disjunction of two propositions, p and q , is written: $p \vee q$ read as "p or q " or "p disjunction q ."**

P	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example

p : 3 is an odd number. (True)

q : 27 is a prime number. (False)

$p \vee q$: 3 is an odd number or 27 is a prime number. (True)

3. Implication

When two propositions are joined with the connective "**implies**," the proposition formed is called a *logical* ***implication*****. And denoted by "$\Longrightarrow$" written: $p \Longrightarrow q$ read as "p implies q ."**

p	q	$p \Longrightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Examples

p : 3 is an odd number. (True)

q : 27 is a prime number. (False)

$p \Longrightarrow q$: If 3 is an odd number, then 27 is prime. (False)

There are various ways of expressing the proposition $p \Longrightarrow q$,

- If p , then q or q if p or p implies q .
- p only if q or p is sufficient for q .
- q is necessary for p

4. Bi-implication

When two propositions are joined with the connective "**bi implication**," the proposition formed is called a *logical* ***bi-implication*** **or a** ***logical equivalence*** **and denoted by "$\Longleftrightarrow$" and written:$p \Longleftrightarrow q$.**

$p \Longleftrightarrow q$ is false if and only if p and q have different truth values.

P	q	$p \Longleftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Examples 1.4:

Let p : 2 is greater than 3. (False)

q : 5 is greater than 4. (True)

Then $p \Longleftrightarrow q$: 2 is greater than 3 if and only if 5 is greater than 4.(False)

There are various ways of stating the proposition $p \Longleftrightarrow q$.

- p if and only if q (also written as p iff q),
- p implies q and q implies p ,
- p is necessary and sufficient for q
- q is necessary and sufficient for p

5. Negation

Given any proposition p , we can form the proposition $\neg p$ called the **negation** of p .
The truth value of $\neg p$ is F if p is T and T if p is F .

P	$\neg p$
T	F
F	T

Example 1.5: Let p : Addis Ababa is the capital city of Ethiopia. (True)
$\neg p$: Addis Ababa is not the capital city of Ethiopia. (False)

Compound (or complex) propositions

Definition The proposition formed by joining two or more proposition by connective(s) is called a compound statement.

The possible truth values of a proposition are often listed in a table, called a **truth table**. If p and q are propositions, then there are four possible combinations of truth values for p and q. That is, TT, TF, FT and FF. If a third proposition r is involved, then there are eight possible combinations of truth values for p,q and r. In general, a truth table involving "n" propositions $p_1, p_2, \ldots, p_n$ contains 2^n possible combinations of truth values for these propositions and a truth table showing these combinations would have n columns and 2^n rows. So, we use truth tables to determine the truth value of a compound proposition based on the truth value of its constituent component propositions.

Remark: When dealing with compound propositions, we shall adopt the following convention on the use of parenthesis. Whenever "$\vee$" or "$\wedge$" occur with "$\Rightarrow$" or "$\Leftrightarrow$", we shall assume that "$\vee$" or "$\wedge$" is applied first, and then "$\Rightarrow$" or "$\Leftrightarrow$" is then applied. For example,

Examples

Suppose that a compound proposition $(p \vee q) \Longrightarrow (r \Longleftrightarrow \neg s)$ and that the truth values of p, q, r, and s are T, F, F, and T, respectively. Then the truth value of $(p \vee q) \Longrightarrow (r \Longleftrightarrow \neg s)$ is T.

Definition Two compound propositions P and Q are said to be ***equivalent*** if they have the same truth value for all possible combinations of truth values for the component propositions occurring in both P and Q. In this case we write $P \equiv Q$.

Let $P : p \Longrightarrow q$.

Q: $\neg q \Longrightarrow p$.

p	q	¬ p	¬ q	$p \Longrightarrow q$	¬ q ⟹ p
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Give proposition $p \Longrightarrow q$, **then**

- The **Converse:** of $p \Longrightarrow q$ is $q \Longrightarrow p$
- The **Contrapositive** of $p \Longrightarrow q$ is $\neg q \Longrightarrow \neg p$
- The **Inverse** of $p \Longrightarrow q$ is $\neg p \Longrightarrow q$

Example: If Kidist lives in Addis Ababa, then she lives in Ethiopia.

Converse: If Kidist lives in Ethiopia, then she lives in Addis Ababa.

Contrapositive: If Kidist does not live in Ethiopia, then she does not live in Addis Ababa.

Inverse: If Kidist does not live in Addis Ababa, then she does not live in Ethiopia.

Law or identities

1. Idempotent Laws

a. $p \equiv p \vee p$.

b. $p \equiv p \wedge p$.

2. Commutative Laws

a. $p \wedge q \equiv q \wedge p$.

b. b. $p \vee q \equiv q \vee p$.

3. Associative Laws

a. $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.

b. $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

4. Distributive Laws

a. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

b. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

5. De Morgan's Laws

a. $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

b. $\neg(p \vee q) \equiv \neg p \wedge \neg q$

6. Complement Law

$\neg(\neg p) \equiv p$.

Definition: A compound proposition is a **tautology** if it is always true regardless of the truth values of its component propositions. If, on the other hand, a compound proposition is always false regardless of its component propositions, we say that such a proposition is a **contradiction**.

Examples Consider the compound propositions $p \vee \neg p$ and $p \wedge \neg p$.

p	$\neg p$	p $\vee \neg p$	p $\wedge \neg p$
T	F	T	F
F	T	T	F

Open propositions and quantifiers

Definition: is a sentence that contains one or more variables and whose truth value depends on the values assigned for the variables.

Examples:

a. x is the day before Sunday.

b. y is a city in Africa.

c. x is greater than y .

d. x + 4 = –9

Definition: Two open proposition $P\ (x\ )$ and $Q\ (x\ )$ are said to be equivalent if and only if $P\ (a\ ) = Q\ (a\ )$ for all individual a .
Note that if the universe U is specified, then $P\ (x\ )$ and $Q\ (x\ )$ are equivalent iff $P\ (a\ ) = Q\ (a\ )$ for all $a \in U$.

Example
Let $P\ (x\ )$: x^2-1= 0. and $Q\ (x\ )$: $|x\ | \geq 1$.

Let U = {−1, − 1/2 ,0,1}. Then for all $a \in U$; $P\ (a\ )$ and $Q\ (a\ )$ have the same truth value.

$P\ (-1)$: $(-1)^2 - 1 = 0\ (T\ )$ & $Q\ (-1)$: $|-1| \geq 1$ T

$P\ (-1/2)$: $(-\frac{1}{2})^2 - 1 = 0\ (F\ )$ & $Q\ (-1/2)$: $|-1/2| \geq 1$ F

$P\ (0)$: $0 - 1 = 0\ (F\ )$ & $Q\ (0)$: $|0| \geq 1$ F

$P\ (1)$: $1 - 1 = 0\ (T\ )$ & $Q\ (1)$: $|1| \geq 1$ T

Therefore $P\ (a\ ) = Q\ (a\ )$ for all $a \in U$

Definition: Let U be the universal set,
An open proposition $P(x)$ is a **tautology** if and only if $P(a)$ is always true for all values of $a \in U$
Example The open proposition $P(x)$: $x^2 \geq 0$ is a tautology.

a. The phrase "for every x " is called a **universal quantifier**. We regard

- for every x ,
- for all x ,
- for each x

as having the same meaning & symbolize each by **$(\forall x)$.**

If $P(x)$ is an open proposition with universe U , then
$(\forall x)P(x)$ is a quantified proposition and is read as "every $x \in U$ has the property P ."

b. The phrase "there exists an x " is called an **existential quantifier**. We regard

- There exists an x ,
- For some x ,
- For at least one x

as having the same meaning, and symbolize each by **"$(\exists x)$."**

If $P(x)$ is an open proposition with universe U, then $(\exists x)P(x)$ is a quantified proposition and is read as "there exists $x \in U$ with the property P .

Remarks:

i. To show that $(\forall x)P(x)$ is F, it is sufficient to find at least one $a \in U$ such that $P(a)$ is F. Such an element $a \in U$ is called a ***counter example***.

ii. $(\exists x)P(x)$ is F if we cannot find any $a \in U$ having the property P.

Example

a. Write the following statements using quantifiers.

i. For each real number $x > 0$, x^2+x-6=0.

Solution: $(\forall x > 0)(x^2$+x-6=0).

ii. There is a real number $x > 0$ such that x^2+x-6=0

Solution: $(\exists x > 0)(x^2$+x-6=0).

Relationship between the existential and universal quantifiers

If $P(x)$ is a formula in x, consider the following four statements.

a. $(\forall x)P(x)$. Everything has property P.

b. $(\exists x)P(x)$. Something has property P

c. $(\forall x)\neg P(x)$. Nothing has property P.

d. $(\exists x)\neg P(x)$. Something does not have property P.

Notes:

$$\neg(\forall x)P(x) \equiv (\exists x)\neg P(x).$$

$$\neg(\exists x)P(x) \equiv (\forall x)\neg P(x).$$

Example Let $U = \mathbb{R}$.

a. $\neg(\exists x)(x < x^2) \equiv (\forall x) \neg(x < x^2)$

$\equiv (\forall x)(x \geq x^2)$.

b. $\neg(\forall x)(4x + 1 = 0) \equiv (\exists x) \neg(4x + 1 = 0)$

$\equiv (\exists x)(4x + 1 \neq 0)$.

a. All rational are reals. $(\forall x)(\mathbb{Q}(x) \Longrightarrow \mathbb{R}(x))$.

b. No rational are reals. $(\forall x)(\mathbb{Q}(x) \Longrightarrow \neg\mathbb{R}(x))$.

c. Some rational are reals. $(\exists x)(\mathbb{Q}(x) \wedge \mathbb{R}(x))$.

d. Some rational are not reals. $(\exists x)(\mathbb{Q}(x) \wedge \neg\mathbb{R}(x))$.

Example Let U = The set of integers.

Let $P(x)$: x is a prime number.

$Q(x)$: x is an even number.

$R(x)$: x is an odd number.

Then

a. $(\exists x)[P(x) \Longrightarrow Q(x)]$ is T; since there is an x, say 2, such that $P(2) \Longrightarrow Q(2)$ is T.

b. $(\forall x)[P(x) \Longrightarrow Q(x)]$ is F.
As a counterexample take 7. Then $P(7)$ is T and $Q(7)$ is F.

Hence $P(7) \Longrightarrow Q(7)$.

c. $(\forall x)[R(x) \wedge P(x)]$ is F.

Quantifiers Occurring in Combinations

1. $(\forall x)(\forall y)P(x, y)$ for all x and for all y the relation $P(x, y)$ holds;

2. $(\exists x)(\exists y)P(x, y)$ there is an x and there is a y for which $P(x, y)$ holds;

3. $(\forall x)(\exists y)P(x, y)$ for every x there is a y such that $P(x, y)$ holds";

4. $(\exists x)(\forall y)P(x, y)$ there is an x which stands to every y in the relation $P(x, y)$."

Example Let U = The set of integers. Let $P(x, y)$: $x + y = 5$.

a. $(\exists x)(\exists y)P(x, y)$ means that there is an integer x and there is an integer y such that $x + y = 5$. This statement is true when $x = 4$ and $y = 1$, since 4 + 1 = 5.

Therefore, the statement $(\exists x)(\exists y)P(x, y)$ is always true for this universe.

b. $(\exists x)(\forall y)P(x, y)$ means that there is an integer x_0 such that for every y, $x_0 + y = 5$. This is false since no fixed value of x will make this true for all y in the universe; e.g. if $x_0 = 1$, then $1 + y = 5$ is false for some y.

c. $(\forall x)(\exists y)P(x, y)$ means that for every integer x, there is an integer y such that $x + y = 5$.
Let $x = a$, then $y = 5 - a$ will always be an integer, so this is a true statement.

d. $(\forall x)(\forall y)P(x, y)$ means that for every integer x and for every integer y, $x + y = 5$. This is false, for if $x = 2$ and $y = 7$, we get $2 + 7 = 9 \neq 5$.

Argument and Validity

Definition An argument (logical deduction) is an assertion that a given set of statements $p\ 1, p\ 2, p\ 3, \ldots, p\ n$, called **hypotheses or *premises***, yield another statement Q , called the *conclusion*. Such a logical deduction is denoted by:

$p\ 1, p\ 2, p\ 3, \ldots, p\ n \vdash Q$ **or**

$p\ 1$

$p\ 2$

.

.p_n

—

Q

Example: If you study hard, then you will pass the exam. You did not pass the exam. Therefore, you did not study hard.

Let

p : You study hard.

q : You will pass the exam.

$p \Longrightarrow q$

$\neg p$

—

$\neg q$

Definition: An argument form $p\ 1, p\ 2, p\ 3, \ldots, p\ n \vdash Q$ is said to be ***valid*** **if** Q **is true whenever all the premises** $p\ 1, p\ 2, p\ 3, \ldots, p\ n$ **are true; otherwise it is *invalid*.**

Example 1.20: Investigate the validity of the following argument:

a. $p \Longrightarrow q, \neg q \vdash \neg p$

b **a.**

p	q	$\neg p$	$\neg q$	$p \Rightarrow q$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

c. If it rains, crops will be good. It did not rain. Therefore, crops were not good.

c. Let p: It rains.

q: Crops are good.

$\neg p$: It did not rain.

$\neg q$: Crops were not good.

The argument form is $p \Rightarrow q, \neg p \vdash \neg q$

Now we can use truth table to test validity as follows:

What tells us that the argument is valid?

Remember that an argument is valid if it is impossible for the premises to be true and the conclusion to be false.
So, we check to see if there is a row on the truth table that has all true premises and a false conclusion.
If there is, then we know the argument is invalid.

The argument form $p_1, p_2, p_3, \ldots, p_n \vdash Q$ is valid iff the statement $(p_1 \wedge p_2 \wedge p_3 \wedge \ldots \wedge p_n) \Longrightarrow Q$ is a tautology.

Rules of inferences

Below we list certain valid deductions called rules of inferences.

1. Modes Ponens

$$\frac{p \qquad p \Longrightarrow q}{q}$$

2. Modes Tollens

$$\frac{\neg q \qquad p \Longrightarrow q}{\neg p}$$

3. Principle of Syllogism

$$\frac{p \Longrightarrow q \qquad q \Longrightarrow r}{p \Longrightarrow r}$$

4. Principle of Adjunction

a. $$\frac{p \qquad q}{p \wedge q}$$

b. $$\frac{q}{p \vee q}$$

5. Principle of Detachment

$$\frac{p \wedge q}{p, q}$$

6. Modes Tollendo Ponens

$$\frac{\neg p \qquad p \vee q}{q}$$

7. Modes Ponendo Tollens

$$\frac{\neg(p \wedge q)}{\begin{array}{l} p \\ \hline \neg q \end{array}}$$

8. Constructive Dilemma

$$\frac{(p \Rightarrow q) \wedge (r \Rightarrow s)}{\begin{array}{l} p \vee r \\ \hline q \vee s \end{array}}$$

9. Principle of Equivalence

$$\frac{p \Leftrightarrow q}{\begin{array}{l} p \\ \hline q \end{array}}$$

10. Principle of Conditionalization

$$\frac{\begin{array}{l} p \\ \hline \end{array}}{q \Rightarrow p}$$

Formal proof of validity of an argument

Definition 1.9: A formal proof of a conclusion Q given hypotheses $p_1, p_2, p_3, \ldots, p_n$ is a sequence of stapes, each of which applies some inference rule to hypotheses or previously proven statements (antecedent) to yield a new true statement (the consequent).

Example 1.21: Show that $p \Rightarrow \neg q, q \vdash \neg p$ is valid.

Solution:

1.	q is true	premise
2.	$p \Rightarrow \neg q$	premise
3.	$q \Rightarrow \neg p$	contrapositive of (2)
4.	$\neg$p	Modes Ponens using (1) and (3)

Set Theory

Description of sets

Sets are described or characterized by one of the following four different ways.

- **Verbal Method**
- **Roster/Complete Listing Method**
- **Partial Listing Method**
- **Set-builder Method**

Definition : The set which has no element is called the empty (or null) set and denoted by ϕ or {}.

Relationships between two sets

Definition Set B is said to be a ***subset*** **of set** A **(or is contained in** A **), denoted by** $B \subseteq A$ **, if every element of** B **is an element of** A **, i.e.,** $(\forall x)(x \in B \Rightarrow x \in A)$**.**

It follows from the definition that set B **is not a subset of set** A **if at least one element of** B **is not an element of** A **. i.e.,** $B \not\subseteq A \Leftrightarrow (\exists x)(x \in B \Rightarrow x \notin$ **?? In such cases we write** $B \not\subseteq A$ **or** $A \not\supseteq B$

Definition Sets A and B are said to be ***equal*** **if they contain exactly the same elements. In this case, we write** $A = B$ **. That is,** $(\forall x)(x \in B \iff x \in A)$.

Example

a. The sets {1, 2, 3}, {2, 1, 3}, {1, 3, 2} are all equal.

b. {x | x is a counting number} = {x | x is a positive integer}

Definition Set A is said to be a **proper subset** of set B if every element of A is also an element of B, but B has at least one element that is not in A. In this case, we write $A \subset B$. We also say B is a proper super set of A, and write $B \supset A$. It is clear that $A \subset B \iff [(\forall x)(x \in A \implies x \in B) \wedge (A \neq B)]$.

Definition Let A be a set. The power set of A, dented by $P(A)$, is the set whose elements are all subsets of A. That is, $P(A) = \{B : B \subseteq A\}$.

Example Let $A = \{x, y, z\}$. As noted before, ϕ and A are subset of A. Moreover, $\{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}$ and $\{y, z\}$ are also subsets of A.
Therefore,
$P(A) = \{\phi, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, A\}$.

Set Operations and Venn diagrams

Given two subsets A and B of a universal set U, new sets can be formed using A and B in many ways, such as taking common elements or non-common elements, and putting everything together. Such processes of forming new sets are called ***set operations***.

In this section, three most important operations, are

- **union**
- **intersection**
- **and complement are discussed.**

Definition The union of two sets A and B, denoted by $A \cup B$, is the set of all elements that are either in A or in B (or in both sets). That is, $A \cup B = \{x : (x \in A) \vee (x \in B)\}$.

Definition The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements that are in A and B. That is, $A \cap B = \{x : (x \in A) \wedge (x \in B)\}$.

Definition The difference between two sets A and B, denoted by $A - B$, is the of all elements in A and not in B; this set is also called the **relative complement** of B with respect to A.
Symbolically, $A - B = \{x : x \in A \wedge x \notin B\}$.

Example If A = {1,3,5}, B = {1,2}, then $A - B$ = {3,5} and $B - A$ = {2}.

Definition Let A be a subset of a universal set U. The *absolute complement* (or simply *complement*) of A, denoted by A' (or A^c or $\bar{A}$), is defined to be the set of all elements of U that are not in A. That is,
$A' = \{x : x \in U \wedge x \notin A\}$ or $x \in A' \iff x \notin A \iff \neg(x \in$?
?

Theorem For any two sets A and B, each of the following holds.

1. $(A')' = A$.
2. $A' = U - A$.
3. $A - B = A \cap B'$.
4. $(A \cup B)' = A' \cap B'$.
5. $(A \cap B)' = A' \cup B'$.
6. $A \subseteq B \iff B' \subseteq A'$

Definition The symmetric difference of two sets A and B, denoted by $A \Delta B$, is the set $A \Delta B = (A - B) \cup (B - A)$.

Theorem For any three sets A , B and C , each of the following holds.

a. $A \cup B = B \cup A$. (is commutative)

b. $A \cap B = B \cap A$. (is commutative)

c. $(A \cup B) \cup C = A \cup (B \cup C)$. (is associative)

d. $(A \cap B) \cap C = A \cap (B \cap C)$. (is associative)

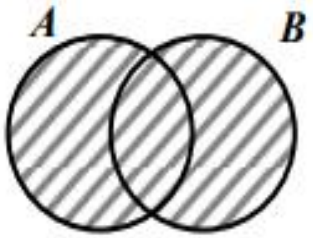
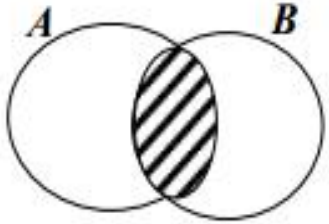
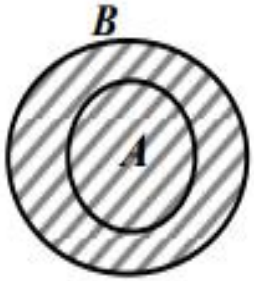
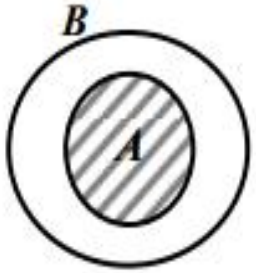
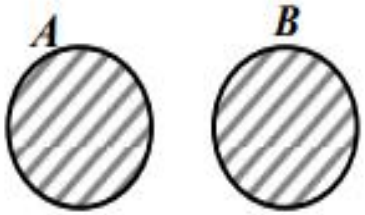
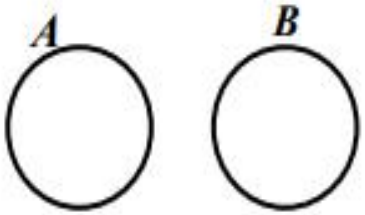
e. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. (is distributive over)

f. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. (is distributive over $\cup$)

Venn diagrams

A Venn diagram is a schematic or pictorial representative of the sets.

Now we illustrate intersections and unions of sets by Venn diagram.

Cases	Shaded is $A \cup B$	Shaded $A \cap B$
Only some common elements		
$A \subseteq B$		
No common element		 $A \cap B = \varnothing$